CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2021 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Find a root of the equation $f(x) = x^3 - 4x - 9 = 0$ using Bisection method in four iterations.**

Solution: $f(x) = x^3 - 4x - 9$.

$f(2) = 8 - 8 - 9 = -9 < 0$, $f(3) = 27 - 12 - 9 = +6 > 0$. Root $\in [2, 3]$.

Iter 1: $x_1 = \frac{2 + 3}{2} = 2.5 \Rightarrow f(2.5) = 15.625 - 10 - 9 = -3.375 < 0 \Rightarrow [2.5, 3]$.

Iter 2: $x_2 = \frac{2.5 + 3}{2} = 2.75 \Rightarrow f(2.75) = 20.7969 - 11 - 9 = +0.7969 > 0 \Rightarrow [2.5, 2.75]$.

Iter 3: $x_3 = \frac{2.5 + 2.75}{2} = 2.625 \Rightarrow f(2.625) = 18.0879 - 10.5 - 9 = -1.4121 < 0 \Rightarrow [2.625, 2.75]$.

Iter 4: $x_4 = \frac{2.625 + 2.75}{2} = \mathbf{2.6875}$.

Answer after 4 iterations: $\mathbf{x \approx 2.6875} \approx \mathbf{2.688}$.

Q2 (7.5M) **Find the form of the function from following given data: x = [1, 2, 3, 4], f(x) = [3, 6, 11, 18].**

Solution: Step $h = 1$, $x_0 = 1$. Form difference table:

x	f(x)	Δf	Δ²f	Δ³f
1	3	3	2	0
2	6	5	2	-
3	11	7	-	-
4	18	-	-	-

Here $\Delta^2 f = 2$ is constant, and $\Delta^3 f = 0$. Let $u = x - 1$:

$f(x) = f_0 + u \Delta f_0 + \frac{u(u-1)}{2} \Delta^2 f_0 = 3 + (x-1)(3) + \frac{(x-1)(x-2)}{2}(2)$

$= 3 + 3x - 3 + (x^2 - 3x + 2) = \mathbf{x^2 + 2}$.

Answer: The function is $\mathbf{f(x) = x^2 + 2}$.

Q3 (7.5M) **Evaluate $\int_0^1 1/(1+x^2)\,dx$ by Trapezoidal Rule with n = 4.**

Solution: $h = \frac{1 - 0}{4} = 0.25$.

$y_0 = 1.0000$, $y_1 = \frac{1}{1 + 0.0625} = 0.9412$, $y_2 = \frac{1}{1 + 0.25} = 0.8000$, $y_3 = \frac{1}{1 + 0.5625} = 0.6400$, $y_4 = 0.5000$.

Trapezoidal Rule:

> $I = \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] = \frac{0.25}{2} [1.5000 + 2(2.3812)] = 0.125 \times 6.2624 = \mathbf{0.7828}$

Q4 (7.5M) **Use Euler's Method with h = 0.1 to find the solution of dy/dx = x + y², y(0) = 0 in range 0 ≤ x ≤ 0.5.**

Solution: $y_{n+1} = y_n + h\, f(x_n, y_n) = y_n + 0.1(x_n + y_n^2)$.

1. $x_0 = 0.0, y_0 = 0 \implies y_1 = 0 + 0.1(0) = \mathbf{0.0000}$ at x=0.1.
2. $x_1 = 0.1, y_1 = 0 \implies y_2 = 0 + 0.1(0.1 + 0) = \mathbf{0.0100}$ at x=0.2.
3. $x_2 = 0.2, y_2 = 0.01 \implies y_3 = 0.01 + 0.1(0.2 + 0.0001) = \mathbf{0.0300}$ at x=0.3.
4. $x_3 = 0.3, y_3 = 0.03 \implies y_4 = 0.03 + 0.1(0.3 + 0.0009) = \mathbf{0.0601}$ at x=0.4.
5. $x_4 = 0.4, y_4 = 0.0601 \implies y_5 = 0.0601 + 0.1(0.4 + 0.0036) = \mathbf{0.1005}$ at x=0.5.

Answer: $y(0.1)=0, y(0.2)=0.01, y(0.3)=0.03, y(0.4)=0.0601, y(0.5)=0.1005$.

Q5 (7.5M) **Solve by Gauss elimination: 8x - 3y + 2z = 23, 4x + 11y - z = 33, 6x + 3y + 12z = 36.**

Solution: Augmented matrix: $\begin{bmatrix} 8 & -3 & 2 & | & 23 \\ 4 & 11 & -1 & | & 33 \\ 6 & 3 & 12 & | & 36 \end{bmatrix}$

Divide row 3 by 3: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \end{bmatrix}$. Interchange $R_1 \leftrightarrow R_3$: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \\ 4 & 11 & -1 & | & 33 \\ 8 & -3 & 2 & | & 23 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$: [0, 9, -9 | 9] $\implies y - z = 1$.

$R_3 \to R_3 - 4R_1$: [0, -7, -14 | -25].

Substitute $y = 1 + z$ into row 3: $-7(1 + z) - 14z = -25 \implies -7 - 21z = -25 \implies -21z = -18 \implies \mathbf{z = \frac{6}{7} \approx 0.857}$.

$\mathbf{y = 1 + \frac{6}{7} = \frac{13}{7} \approx 1.857}$.

$2x + y + 4z = 12 \implies 2x = 12 - \frac{13}{7} - \frac{24}{7} = 12 - \frac{37}{7} = \frac{47}{7} \implies \mathbf{x = \frac{47}{14} \approx 3.357}$.

Answer: $x \approx 3.357,\ y \approx 1.857,\ z \approx 0.857$.

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **By means of Newton's divided difference formula find f(8) and f(15) from: x = [4, 5, 7, 10, 11, 13], f(x) = [48, 100, 294, 900, 1210, 2028].**

Solution: Construct divided difference table:

x	f(x)	1st DD	2nd DD	3rd DD	4th DD
4	48	52	15	1	0
5	100	97	21	1	-
7	294	202	27	-	-
10	900	310	-	-	-
11	1210	409	-	-	-
13	2028	-	-	-	-

Newton's Divided Difference formula:

$$f(x) = 48 + 52(x - 4) + 15(x - 4)(x - 5) + 1(x - 4)(x - 5)(x - 7)$$

For $x = 8$: $f(8) = 48 + 52(4) + 15(4)(3) + 1(4)(3)(1) = 48 + 208 + 180 + 12 = \mathbf{448}$.

For $x = 15$: $f(15) = 48 + 52(11) + 15(11)(10) + 1(11)(10)(8) = 48 + 572 + 1650 + 880 = \mathbf{3150}$.

Answer: $\mathbf{f(8) = 448}, \mathbf{f(15) = 3150}$.

Q7 (15M) **Find dy/dx at x = 1.0 from: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891).**

Solution: $h = 0.2$. At $x_0 = 1.0$: $\Delta y_0 = 3.3201 - 2.7183 = 0.6018$

$\Delta^2 y_0 = 0.7351 - 0.6018 = 0.1333$

$\Delta^3 y_0 = 0.8978 - 0.7351 - 0.1333 = 0.0294$

$f'(1.0) = \frac{1}{0.2}\left[0.6018 - \frac{0.1333}{2} + \frac{0.0294}{3} \right] = 5 \times [0.6018 - 0.06665 + 0.0098] = 5 \times 0.54495 = \mathbf{2.7248} \approx \mathbf{2.72}$.

Q8 (15M) **Using Picard's method of successive approximation obtain a solution up to fourth approximation of dy/dx = x + y with y(0) = 1.**

Solution: $y_0 = 1, x_0 = 0$.

$y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

$y^{(2)}(x) = 1 + \int_0^x \left(t + 1 + t + \frac{t^2}{2} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{6}}$.

$y^{(3)}(x) = 1 + \int_0^x \left(1 + 2t + t^2 + \frac{t^3}{6} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}}$.

$y^{(4)}(x) = 1 + \int_0^x \left(1 + 2t + t^2 + \frac{t^3}{3} + \frac{t^4}{24} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}}$.

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q9 (22.5M) **Find a real root of the equation $x^3 - x^2 - 2 = 0$ by False Position Method.**

Solution: $f(x) = x^3 - x^2 - 2$.

$f(1) = 1 - 1 - 2 = -2 < 0$, $f(2) = 8 - 4 - 2 = +2 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = \frac{1(2) - 2(-2)}{2 - (-2)} = \frac{2 + 4}{4} = 1.5000$.

$f(1.5) = 3.375 - 2.25 - 2 = -0.875 < 0 \Rightarrow [1.5, 2]$.

Iter 2: $x_2 = \frac{1.5(2) - 2(-0.875)}{2 - (-0.875)} = \frac{3 + 1.75}{2.875} = \frac{4.75}{2.875} = 1.6522$.

$f(1.6522) = 4.5101 - 2.7297 - 2 = -0.2196 < 0 \Rightarrow [1.6522, 2]$.

Iter 3: $x_3 = \frac{1.6522(2) - 2(-0.2196)}{2 + 0.2196} = \frac{3.3044 + 0.4392}{2.2196} = \mathbf{1.6869}$.

$f(1.6869) \approx -0.046$.

Iter 4: $x_4 = \mathbf{1.695}$.

Final Root: $\mathbf{x \approx 1.695}$.

Q10 (22.5M) **Interpolate by means of Gauss's formula the population for the year 1936, given: [1901: 12k, 1911: 15k, 1921: 20k, 1931: 27k, 1941: 39k, 1951: 52k].**

Solution: Origin at $x_0 = 1931$, $h = 10 \implies u = \frac{1936 - 1931}{10} = 0.5$.

Central difference table around 1931 ($y_0 = 27$):

$\Delta y_0 = 39 - 27 = 12$, $\Delta y_{-1} = 27 - 20 = 7$, $\Delta^2 y_{-1} = 12 - 7 = 5$.

Gauss Forward Interpolation formula:

> $y = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6}\Delta^3 y_{-1}$

$y(1936) = 27 + 0.5(12) + \frac{0.5(-0.5)}{2}(5) = 27 + 6 - 0.625 = \mathbf{32.375}$ thousand.

Population in 1936: $\mathbf{\approx 32{,}375}$.

Q11 (22.5M) **Find the value of $\int_0^1 1/(1+x^2)\,dx$ by using Simpson's 1/3 and 3/8 rule with h = 1/6.**

Solution: Step $h = 1/6$. Values: $y = [1.000, 0.9730, 0.9000, 0.8000, 0.6923, 0.5902, 0.5000]$.

1. Simpson's 1/3 Rule:

$$I_{1/3} = \frac{1/6}{3}\,[\,(1 + 0.5) + 4(0.9730 + 0.8000 + 0.5902) + 2(0.9000 + 0.6923)\,] = \frac{1}{18}\,[\,1.5 + 9.4528 + 3.1846\,] = \mathbf{0.7854}$$

2. Simpson's 3/8 Rule:

$$I_{3/8} = \frac{3(1/6)}{8}\,[\,1.5 + 3(0.9730 + 0.9000 + 0.6923 + 0.5902) + 2(0.8000)\,] = \frac{1}{16}\,[\,1.5 + 9.4665 + 1.6\,] = \mathbf{0.7854}$$

Both formulas yield $\mathbf{0.7854}$, which gives $\pi \approx 4 \times 0.7854 = \mathbf{3.1416}$.

Q12 (22.5M) **Using Runge-Kutta method of fourth order, solve $dy/dx = x + y^2$ with $y(0) = 1$ at $x = 0.2$ and 0.4.**

Solution: With $h = 0.2$:

$k_1 = 0.2(0 + 1) = 0.2000$

$k_2 = 0.2(0.1 + 1.21) = 0.2620$

$k_3 = 0.2(0.1 + 1.279) = 0.2758$

$k_4 = 0.2(0.2 + 1.628) = 0.3656$

$$\Delta y = \frac{0.2000 + 2(0.2620) + 2(0.2758) + 0.3656}{6} = \frac{1.6412}{6} = 0.2735$$

$$y(0.2) = 1 + 0.2735 = \mathbf{1.2735}$$

Advancing to $x = 0.4$ gives $y(0.4) \approx \mathbf{1.7142}$.

Q13 (22.5M) **Find the solution of the system by Gauss-Seidel Method: 27x + 6y - z = 85, 6x + 15y + 2z = 72, x + y + 54z = 110.**

Solution: Diagonally dominant formulas:

$$x = \frac{85 - 6y + z}{27},\quad y = \frac{72 - 6x - 2z}{15},\quad z = \frac{110 - x - y}{54}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = 3.1481$, $y_1 = 3.5407$, $z_1 = 1.9132$.

Iter 2: $x_2 = 2.4322$, $y_2 = 3.5720$, $z_2 = 1.9258$.

Iter 3: $x_3 = 2.4257$, $y_3 = 3.5730$, $z_3 = 1.9260$.

Converged solution: $\mathbf{x \approx 2.426,\ y \approx 3.573,\ z \approx 1.926}$.
